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Cisco Secure Workload — GxP Compliance Framework

Reference Design | Life Sciences Accounts (Pharma · Biotech · Medical Devices · CDMO/CRO)

Version: 1.0 **Audience:** Solutions Architects, Security Engineers **Use Case:** Fresh Install, Hybrid Environment (On-Prem + Cloud + Manufacturing OT) **Regulatory basis:** EU GMP Annex 11 · FDA 21 CFR Part 11 · GAMP 5 (2nd Ed.) · FDA Computer Software Assurance (2022 draft guidance)

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1. Overview

This reference design guides deployment of Cisco Secure Workload (CSW) in regulated life-sciences environments to produce **inspection-ready evidence** for GxP computerised-system controls.

GxP is **not a certification**. There is no GxP certificate, no attestation report, and no audit opinion. Customers are inspected by regulators (FDA, EMA, MHRA) and must demonstrate a **validated state** on demand. CSW's role is to make a subset of those controls *enforced and evidenced* rather than asserted in a document.

What GxP means here

Regulation	Scope	Why CSW matters
EU GMP Annex 11	Computerised systems used in GMP activities	Clauses 9 (audit trails), 10 (change control), 12 (security), 16 (business continuity)
21 CFR Part 11	Electronic records and signatures (US)	§11.10(d) access limitation, §11.10(e) audit trails, §11.10(k) documentation controls
GAMP 5 (2nd Ed.)	Risk-based validation methodology	Infrastructure qualification; supplier assessment
ALCOA+	Data-integrity principles	CSW evidences *control* integrity, not record integrity — see §1.2

1.1 Where CSW genuinely contributes

GxP need	CSW capability
Segregation of validated from non-validated systems	Scopes, labels, enforced segmentation policy with flow evidence
Logical access restriction to authorised systems	Workload-identity policy (not IP-based ACLs)
Change control on the validated environment	Policy versioning, simulate-before-enforce, drift detection
Audit trail of the security control	Immutable flow records, policy change history
Compensating controls for un-patchable validated systems	Native CVE inventory + targeted segmentation. **This is the strongest single use case — see §8**
Incident reconstruction	Full flow and process forensic telemetry

1.2 Scope boundary — state this before the customer asks

CSW does **not**:

- Perform Computer System Validation. It cannot validate an application, and it is not a CSV/CSA tool.
- Produce **ALCOA+ evidence at the record level**. CSW never touches the GxP record itself — the batch record, LIMS result, chromatography data file or clinical dataset. Its audit trail is of the *security control*, not the *regulated record*. A quality auditor will make this distinction immediately.
- Provide **electronic signature** capability. All of Part 11 Subpart C (§11.50, §11.70, §11.100–§11.300) is out of scope.
- Qualify infrastructure on the customer's behalf (IQ/OQ/PQ remains theirs).

> **Expect the reverse question.** If CSW is deployed into a GxP environment, the customer's quality organisation may require **CSW itself to be qualified** as a supporting system. Be ready for a supplier assessment, a request for qualification documentation, and for CSW upgrades to fall under their change control. Raise this proactively — being first to say it builds more credibility than any feature.

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2. Pre-Deployment Checklist

- □ CSW cluster (SaaS or on-prem) provisioned and accessible
- □ Network connectivity from workloads to CSW cluster (port 443 outbound)
- □ Linux/Windows agent compatibility verified — **including legacy OS on validated systems** (see §11)
- □ Cloud provider accounts connected (AWS, Azure, GCP) via CSW cloud connectors

- ☐ **Validated System Inventory (VSI) obtained** from the quality organisation — this is the authoritative GxP scope list, not an IT asset list
- ☐ Stakeholders identified: **QA / Quality Systems**, Validation Lead, CSIRT/Security, Manufacturing IT/OT, CISO
- ☐ **Change control record raised** for agent installation on validated systems — agent install on a validated system is itself a change
- ☐ Agreement on whether CSW is in scope for qualification (see §1.2)
- ☐ GxP vs non-GxP boundary agreed in writing with QA

> **Do not skip the change-control item.** Installing software on a validated system without a change record is a finding in its own right, regardless of what the software does.

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3. Phase 1 — Sensor Deployment (Days 1–10)

Longer than the HIPAA equivalent by design: validated systems move through change control, not change windows.

3.1 On-Premises Workloads

```
# Linux (RHEL/CentOS/Ubuntu)
rpm -ivh tet-sensor-<version>.rpm     # RHEL/CentOS
dpkg -i tet-sensor-<version>.deb      # Ubuntu/Debian

# Verify sensor is running
systemctl status tetd
```

Key Sensor Configuration:

- Enforcement Mode: **Monitoring Only** — never enforce on a validated system before simulation review and QA sign-off
- Data Collection: enable process hash, network flow, vulnerability scan
- Tags from Day 1: `env:production`, `compliance:gxp`, `gxp:validated`, `system:<VSI-ID>`

> **Tag with the Validated System Inventory ID.** The `system:<VSI-ID>` label is what lets you produce evidence the quality organisation can reconcile against their own register. Without it, your reports and their register are two unrelated documents.

3.2 Manufacturing / OT Workloads

GMP manufacturing sits in the Purdue model, and the segmentation story maps cleanly onto ISA/IEC 62443:

Purdue Level	Typical systems	CSW approach
Level 4	ERP (SAP), enterprise IT	Standard agent
Level 3.5	DMZ	Policy boundary — the

Purdue Level	Typical systems	CSW approach
		critical enforcement point
Level 3	MES, historians (OSIsoft PI, Aspen IP.21), site servers	Agent where supported
Level 2	SCADA / HMI (DeltaV, PCS 7, Experion)	Agent only if vendor-supported; otherwise agentless
Levels 0–1	PLCs, sensors, instruments	**No agent.** Visibility via connectors and flow telemetry only

For Levels 0–2, use agentless discovery and treat CSW as the enforcement boundary *around* the cell rather than inside it.

3.3 Cloud Workloads (AWS / Azure / GCP)

Option A — Software Sensor on Cloud VMs: as above.

Option B — CSW Cloud Connectors (agentless):

```
CSW UI → Platform → External Orchestrators
  → Add AWS/Azure/GCP connector
  → Provide IAM Role (AWS) or Service Principal (Azure)
  → Enable VPC Flow Log ingestion
```

3.4 Sensor Validation

```
CSW UI → Manage → Agents
  → Confirm all sensors show "Active"
  → Verify telemetry flowing
  → Check agent version alignment
```

GxP addition: export the agent inventory and reconcile against the VSI. Any validated system *without* an agent is a documented gap requiring either an agent or a compensating control — not silence.

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4. Phase 2 — Scope & Inventory Design (Days 11–18)

4.1 GxP Scope Architecture

Design scopes around **validation status first, application second.** This is the inverse of most deployments, and it is what makes the Annex 11 §12 story work.

```
Root Scope
└── LifeSciences-Org
    ├── GxP-Validated (strict change control)
```

```
    │   ├── Manufacturing-MES
    │   ├── LIMS-Laboratory
    │   ├── CDS-Chromatography        (Empower, Chromeleon, OpenLab)
    │   ├── QMS-EDMS                  (Veeva Vault, TrackWise,
MasterControl)
    │   ├── Historian-EMS-BMS         (environmental / cleanroom
monitoring)
    │   └── Clinical-Systems          (CTMS, EDC, eTMF,
Pharmacovigilance)
    ├── GxP-Supporting (qualified infrastructure)
    │   ├── Validated-Databases
    │   ├── Backup-Archive
    │   └── Time-Sync-NTP             (Annex 11 §9 — audit trail
timestamps)
    ├── Non-GxP-Corporate (standard controls)
    │   ├── Active-Directory
    │   ├── Email-Collaboration
    │   └── General-IT
    └── External-Partners
        ├── CRO-CDMO-Access
        └── Vendor-Remote-Support     (instrument OEM support paths)
```

> **The `Non-GxP-Corporate` → `GxP-Validated` boundary is the single most valuable policy in this entire reference design.** It is the control that Annex 11 and GAMP both expect and that most customers can only assert on paper.

4.2 Inventory Filters for GxP System Discovery

```
Filter: GxP-Candidates-Laboratory
  - Process contains: "empower", "chromeleon", "openlab", "labware",
                      "starlims", "samplemanager", "unicorn"
  - Port listened: 1521 (Oracle), 1433 (MSSQL)
  - Hostname contains: "lims", "cds", "qc", "lab"

Filter: GxP-Candidates-Manufacturing
  - Process contains: "opcenter", "pharmasuite", "deltav", "pcs7",
"pi-server"
  - Port listened: 4840 (OPC UA), 5450 (OSIsoft PI), 502 (Modbus),
                   44818 (EtherNet/IP), 135 (OPC DA / DCOM)
  - Hostname contains: "mes", "scada", "hmi", "historian"

Filter: GxP-Candidates-Quality
  - Process contains: "veeva", "trackwise", "mastercontrol",
"documentum"
  - Hostname contains: "qms", "edms", "dms"

Filter: Legacy-Unpatchable
  - OS version: Windows Server 2008/2012, RHEL 6
  - AND scope: GxP-Validated
  → Feeds the compensating-control workflow in §8
```

4.3 Labelling Strategy

Label Key	Example Values	Purpose
`gxp`	validated, supporting, none	**Primary segregation axis**
`system`	VSI-0142, VSI-0291	Reconciles to the quality register
`gxp_category`	gmp, glp, gcp, gdp, gvp	Which GxP discipline applies
`gamp`	cat3, cat4, cat5	GAMP 5 software category — drives validation rigour
`validation_state`	validated, revalidation-due, retired	Lifecycle tracking
`patch_constraint`	revalidation-required, vendor-locked, none	**Drives §8 compensating controls**
`env`	production, qual, dev	Environment segregation
`owner`	qa-ops, mfg-it, lab-ops	Accountability

5. Phase 3 — Application Dependency Mapping (Days 19–40)

5.1 ADM Configuration

```
CSW UI → Investigate → Application Dependency Mapping
  → New Workspace
  → Name: "GxP-Validated-ADM"
  → Select Scope: GxP-Validated
  → Time Window: 4–6 weeks
  → Start ADM
```

> **Run longer than you would elsewhere — 4 to 6 weeks minimum.** GxP environments have cycles that shorter windows miss entirely: monthly batch release, quarterly stability testing, periodic review, annual product quality review, instrument calibration schedules. A two-week ADM in a pharma plant will generate policy that breaks at month-end.

Collection settings:

- Enable **process context**
- Enable **user context** where OS sensors support it
- Retention: align to the customer's GxP record retention, which is commonly **years, not months** — confirm with QA before assuming

5.2 ADM Analysis for GxP

Question	GxP relevance
Which non-GxP systems connect into GxP-Validated?	**Annex 11 §12.1** — the core segregation finding
Do validated systems reach the internet directly?	Annex 11 §12 · uncontrolled change vector
Which vendor/OEM remote-support paths are live?	Annex 11 §3 — supplier access control
Are GxP database connections encrypted?	Annex 11 §7.1 — data storage security
Do all validated systems reach a common NTP source?	**Annex 11 §9** — audit-trail timestamps require reliable time
Unexpected lateral flows inside GxP-Validated?	Annex 11 §12.2 · §13 incident management
Which systems write to the archive/backup tier?	Annex 11 §7.2 · §17 archiving

> The **NTP question is a genuine differentiator.** Annex 11 §9 audit trails and Part 11 §11.10(e) time-stamped records both depend on trustworthy time, and almost nobody can evidence that every validated system actually reaches an approved time source. CSW can show it in a flow query.

5.3 GxP System Identification Workflow

1. **Start from the VSI**, not from ADM output — the quality register is authoritative
2. **Export ADM clusters** and reconcile against the VSI
3. **Flag discrepancies in both directions** — VSI entries with no observed traffic (retired but not decommissioned?) and active clusters absent from the VSI (shadow GxP system?)
4. **Escalate unregistered candidates to QA.** An unregistered system handling GxP data is a finding you have just discovered for them
5. **Tag confirmed systems** with `gxp` and `system:<VSI-ID>`
6. **Re-run ADM** scoped to GxP-Validated for policy generation

> Step 3 is often the first tangible value the customer sees, and it arrives before any policy is written.

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6. Phase 4 — Policy Development (Days 41–60)

6.1 GxP-Aligned Policy Framework

Absolute Policies (always enforced):

```
DENY:  Non-GxP-Corporate → GxP-Validated   (except approved jump
hosts)
DENY:  GxP-Validated → Internet            (no direct egress from
```

```
validated systems)
DENY:  External-Partners → GxP-Validated   (except brokered vendor-
support paths)
DENY:  GxP-Validated → GxP-Validated       cross-system, unless ADM-
confirmed
```

Allowlist Policies (least privilege):

```
ALLOW: LIMS-Laboratory ↔ Validated-Databases      (1521, 1433)
ALLOW: CDS-Chromatography → LIMS-Laboratory       (443)
ALLOW: Manufacturing-MES ↔ Historian              (5450 PI, 4840 OPC
UA)
ALLOW: Manufacturing-MES → ERP                    (3200-3299 SAP)
ALLOW: GxP-Validated → Time-Sync-NTP              (123)   # Annex 11
§9
ALLOW: GxP-Validated → Backup-Archive             (per backup vendor)
# §7.2 · §17
ALLOW: Jump-Host → GxP-Validated                  (22, 3389)
ALLOW: Vendor-Remote-Support → <named system>     (time-bounded,
brokered only)
```

Catch-All (audit everything else):

```
LOG: Any → GxP-Validated     (unmatched inbound — alert)
LOG: GxP-Validated → Any     (unmatched outbound — alert)
```

6.2 Policy Workspace Setup

```
CSW UI → Defend → Segmentation
  → New Workspace
  → Name: "GxP-Validated-Enforcement"
  → Scope: GxP-Validated
  → Import ADM policies (auto-generated baseline)
  → Review and refine with QA + system owners
  → Set mode: Simulation first → then Enforcement
```

6.3 Enforcement Progression

Phase	Mode	Duration	Purpose	QA gate
1	Simulation	**4 weeks**	Validate no false positives across a full business cycle	Simulation report reviewed
2	Enforce — non-GxP boundary only	2 weeks	Enforce the Corporate → Validated deny first	Change record approved
3	Enforce — intra-GxP	2–4 weeks	Staged per system	Change record per

Phase	Mode	Duration	Purpose	QA gate
	allowlist			system
4	Full enforcement	Ongoing	Steady-state posture	Periodic review per Annex 11 §11

Each enforcement step is a change to a validated environment and needs its own change-control record. Build the QA gate into the project plan from the start; discovering it at phase 2 costs weeks. Simulation output is the evidence package that gets the change approved — which makes simulate-before-enforce a *regulatory* feature here, not just a safety feature.

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7. Phase 5 — GxP Control Mapping

7.1 EU GMP Annex 11

Annex 11 Clause	Requirement	CSW Implementation	Coverage
§3 Suppliers & Service Providers	Assess and control supplier access	Brokered, time-bounded vendor-support policy; flow evidence of every OEM session	Partial
§4 Validation	System validated for intended use	*Customer-owned.* CSW supplies qualification docs for its own assessment	Supporting
§7.1 Data Storage	Data secured against damage/unauthorised access	Segmentation of data tier; detection of unencrypted paths	Partial
§7.2 Backups	Regular backup, integrity checked	Policy permitting and evidencing backup paths	Partial
§9 Audit Trails	Secure, time-stamped record of GMP-relevant changes	Immutable flow records; policy change history. **Evidence of NTP reachability for all validated systems**	Partial — control layer only
§10 Change &	Changes made in	Policy versioning,	**Strong**

Annex 11 Clause	Requirement	CSW Implementation	Coverage
Configuration Mgmt	a controlled manner	simulate-before-enforce, **out-of-sync drift detection**	
§11 Periodic Evaluation	Systems periodically evaluated as still valid	Scheduled ADM re-runs; policy drift reports; quarterly posture export	**Strong**
§12.1 Security	Physical/logical controls restrict access to authorised persons	**Enforced workload-identity segmentation** — the core control	**Strong**
§12.2 Security	Extent of controls depends on criticality	Scope-tiered policy: Validated / Supporting / Non-GxP	**Strong**
§12.3 Security	Record creation/change/cancellation of access authorisations	Policy change history with approver and timestamp	Partial
§13 Incident Management	Incidents assessed, root cause identified	Forensic flow and process telemetry for reconstruction	**Strong**
§14 Electronic Signature	—	**Out of scope**	None
§16 Business Continuity	Continuity of system in event of breakdown	Agents continue enforcing cached policy if the control plane is unreachable	Partial
§17 Archiving	Data archived, readable, integrity-checked	Policy permitting and evidencing archive paths	Partial

7.2 FDA 21 CFR Part 11 — Subpart B (§11.10 Closed Systems)

Control	Requirement	CSW Implementation	Coverage
§11.10(a)	Validation of systems	*Customer-owned.* CSW is a supporting system	Supporting
§11.10(c)	Protection of records through retention period	Segmentation of record and archive tiers	Partial
§11.10(d)	**Limiting system access to authorised individuals**	**Enforced network-layer access restriction by workload identity**	**Strong**
§11.10(e)	Secure, computer-generated, time-stamped audit trails	Flow and policy audit trail *of the control*; NTP path evidence	Partial — control layer only
§11.10(g)	Authority checks	Policy enforcement that only authorised systems may interact	Partial
§11.10(h)	Device checks — validity of data source/input device	Workload identity (OS fingerprint, process hash) rather than spoofable IP	Partial
§11.10(k)	Documentation controls / change-record audit trail	Policy versioning and change history	Partial
§11.30	Open systems — additional controls	Detection of unencrypted flows; egress restriction	Partial
Subpart C	Electronic signatures (§11.50–§11.300)	**Out of scope**	None

Use the Coverage column honestly in customer materials. "Strong" on four or five controls that the customer currently cannot evidence at all is a better position than claiming broad coverage that a quality auditor will dismantle in ten minutes.

8. Phase 6 — Vulnerability Management and the Un-Patchable System

This is the highest-value section of the reference design for a life-sciences buyer.

8.1 Why this is different from every other vertical

In a GxP environment, patching a validated system is not a maintenance task — it is a **change requiring impact assessment and, frequently, re-validation**. The practical result is that known vulnerabilities sit unremediated on critical systems for months or years, often on operating systems well past end of support, because the instrument vendor only qualifies a specific OS build.

The quality organisation knows this. It is a standing risk item they cannot close. **A documented, enforced compensating control is the answer they have been unable to produce.**

8.2 Vulnerability Detection

```
CSW UI → Investigate → Vulnerability Report
  → Scope: GxP-Validated
  → Filter: CVSS ≥ 7.0
  → Cross-filter: patch_constraint = revalidation-required | vendor-
locked
  → Export for the quality record
```

8.3 Risk-Based Prioritisation

Prioritise by **exposure**, not CVSS alone. A critical CVE on a system reachable only from one jump host is a lower real risk than a high CVE on a system reachable from the corporate network.

```
Priority 1: Critical CVE + reachable from Non-GxP or External  →
immediate segmentation
Priority 2: Critical CVE + GxP-internal reachability only      →
segment, schedule patch
Priority 3: High CVE + patchable without revalidation          →
normal change control
Priority 4: High CVE + vendor-locked                           →
documented compensating control
```

8.4 The Compensating-Control Workflow

For each vulnerability that cannot be patched within the risk-based timeline:

1. **Document the constraint** — revalidation required, vendor-locked OS, instrument qualification
2. **Reduce reachability** — restrict the vulnerable port to named source workloads only
3. **Enforce it**, don't just alert
4. **Add a forensic rule** on the affected process
5. **Export the evidence**: the CVE, the constraint, the enforced policy, and the flow records proving the restriction holds

6. **File into the quality system** as the compensating control for that risk item

The deliverable is a sentence the customer could not previously write: *"CVE-XXXX-YYYY remains unpatched on VSI-0142 pending revalidation. Reachability is restricted by enforced policy to two named hosts, continuously monitored, with flow evidence retained."* That closes an open risk item, which is worth more to a QA director than any dashboard.

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9. Phase 7 — Monitoring & Alerting

9.1 GxP-Required Alerts

Alert	Trigger	Control
Non-GxP → Validated access	Any unapproved corporate source → GxP-Validated	Annex 11 §12.1 · Part 11 §11.10(d)
Validated system egress to internet	GxP-Validated → external	Annex 11 §12
Unbrokered vendor access	External-Partners → GxP-Validated outside an approved window	Annex 11 §3
NTP path failure	Validated system stops reaching approved time source	**Annex 11 §9 · Part 11 §11.10(e)**
Policy drift / out of sync	Enforced policy no longer matches intended state	**Annex 11 §10**
New lateral flow inside GxP	Previously unobserved east-west flow	Annex 11 §12.2 · §13
Unencrypted GxP data flow	Cleartext protocol from validated system	Annex 11 §7.1
New critical CVE on validated system	CVE appears on `gxp:validated` workload	Annex 11 §1 risk management
Agent offline on validated system	Sensor stops reporting	Annex 11 §9 — evidence gap
Archive/backup path failure	Expected backup flow absent	Annex 11 §7.2 · §17

"Agent offline" is a GxP alert, not just an ops alert. A gap in the evidence record is itself a compliance problem — you cannot demonstrate a control was operating during a period you have no data for.

9.2 Forensic Telemetry for Incident Management (Annex 11 §13)

```
CSW UI → Investigate → Flow Search
  → Time range: incident window
  → Source/Destination: affected validated system
  → Export: full flow log with process context

CSW UI → Investigate → Process Search
  → Filter: GxP workload + time range
  → Identify anomalous processes + parent-child relationships
```

Retain forensic exports to the customer's **GxP record retention period** — confirm with QA. This is commonly far longer than security-industry norms, and in some cases extends to the product lifetime plus a statutory margin.

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10. Reporting & Evidence Collection

10.1 Evidence per Review Cycle

Evidence Item	CSW Source	Annex 11 / Part 11
GxP segregation proof — no unapproved Non-GxP → Validated flows	Flow Search + policy analysis	§12.1 · §11.10(d)
Enforced policy export with version and approver	Defend → Policy Analysis	§10 · §11.10(k)
Policy drift report vs approved baseline	Policy comparison	§10 · §11
Vulnerability report on validated systems	Investigate → Vulnerability	§1 risk management
Compensating-control register	§8.4 workflow output	§1 · §12.2
Scope membership snapshot reconciled to VSI	Inventory → Export	§4 · §11
NTP reachability evidence for all validated systems	Flow Search (port 123)	§9 · §11.10(e)
Vendor/OEM access log	Flow Search (External-Partners)	§3
Agent coverage report vs VSI	Manage → Agents export	§9
Incident reconstruction package	Flow + Process Search	§13

10.2 Ongoing Compliance Posture

- **Monthly** — scope/inventory reconciliation against the VSI; agent coverage check
- **Quarterly** — policy workspace review; drift report; compensating-control register refresh
- **Per periodic review cycle (Annex 11 §11)** — full evidence pack, aligned to the customer's own periodic-evaluation schedule rather than a security calendar
- **Every 90 days** — ADM re-run to detect new dependencies
- **On every validated-system change** — confirm policy still reflects intended state

Align the reporting calendar to **their periodic review schedule**, not to a generic quarterly security cadence. Evidence that arrives in the right format at the right point in the quality cycle gets used; evidence that arrives on a security team's timetable gets filed.

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11. Common Pitfalls

Pitfall	Mitigation
Agent installed without a change record	Raise change control *before* deployment on any validated system. This is the most common own-goal
Claiming "CSW makes you GxP compliant"	There is no such thing. Position as evidence for specific clauses; name the gaps first
Conflating control audit trail with record audit trail	Be explicit that CSW evidences the security control, never the GxP record itself
Legacy OS on validated systems won't take an agent	Plan agentless coverage and boundary enforcement; document as a known gap with compensating control
ADM window too short	Minimum 4–6 weeks. Must span month-end batch release and periodic testing cycles
Enforcement breaks a manufacturing or lab system	Simulation first, always. Stage by scope. Never enforce into a running batch
Scoping from the IT asset list	The **Validated System Inventory from QA** is authoritative. IT asset lists miss instruments and include retired systems
Treating OT like IT	Levels 0–2 often cannot take agents. Enforce at the boundary, observe inside
Retention set to security norms	GxP retention is often years. Confirm

Pitfall	Mitigation
	with QA before configuring
Ignoring whether CSW itself needs qualification	Ask early; supply the supplier assessment pack proactively

12. Product-Accuracy Guardrails

Before any of this goes into customer-facing material:

- **Live Protect is GA for Nexus switches. Live Protect for applications is not GA** — it arrives in CSW 4.1 via the CSW unified agent with Tetragon. Do not present application-level CVE shielding as available today. The §8 compensating-control story stands on **segmentation plus native CVE**, both GA in CSW 4.0, and does not need Live Protect to work.
- **Cisco SmartSwitch (N9300) is a separate Cisco product**, not a CSW feature. It requires its own on-prem VM-based controller, and the service shipping on it today is Hypershield. Do not imply a shared policy plane with CSW.
- **CSW 4.0 GA** covers ADM/AI policy, native CVE, simulation, agents and connectors, Secure Firewall CVE→IPS, and the NVIDIA BlueField DPU agent.

Document prepared for Cisco life-sciences account engagements. Replace [Customer Name] and bracketed fields before customer delivery. GxP is a regulatory regime, not a certification — nothing in this reference design should be presented as conferring compliance.